

Seat pan and backrest pressure distribution while sitting in office chairs



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ABSTRACT

Nowadays, an increasing amount of time is spent seated, especially in office environments, where sitting comfort and support are increasingly important due to the prevalence of musculoskeletal disorders. The aim of this study was to develop a methodology for chair-specific sensor mat calibration, to evaluate the interconnections between specific pressure parameters and to establish those that are most meaningful and significant in order to differentiate pressure distribution measures between office chairs.

The shape of the exponential calibration function was highly influenced by the material properties and geometry of the office chairs, and therefore a chair-specific calibration proved to be essential. High correlations were observed between the eight analysed pressure parameters, whereby the pressure parameters could be reduced to a set of four and three parameters for the seat pan and the backrest respectively. In order to find significant differences between office chairs, gradient parameters should be analysed for the seat pan, whereas for the backrest almost all parameters are suitable.

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1. Introduction

Nowadays, working in an office chair is ubiquitous. Indeed, approximately three-quarters of all employees in industrialised countries now have jobs that require working in a sitting position (Reinecke et al., 2002; Treaster and Marras, 1987). Prolonged seated working presents several problems for physical health as it increases the risk of suffering musculoskeletal disorders in the back, neck, shoulders, arms and legs (Naqvi, 1994; Winkel and Jorgensen, 1986). Despite the general opinion that extended periods of sitting can lead to back complaints, current literature suggests that a sedentary lifestyle by itself does not increase the risk of LBP (Hartvigsen et al., 2000; Kwon et al., 2011; Lis et al., 2007; Roffey et al., 2010). According to May and Lomas (2010) not finding a connection between sitting and LBP is caused by the insidious nature of back pain, since LBP is a very multifactorial condition that can hardly be localised precisely. Recent research also puts the risk of sedentary lifestyle in another perspective. According to Commissaris et al. (2014), physical inactivity is associated with cardiovascular disorders, type II diabetes, depression, obesity as well as with some forms of cancer and 3.2 million people die a

premature death due to an inactive work style. Indeed, every two hours per day increment in sitting time at work increases the risk of obesity by 5% and the risk of diabetes by 7% in female workers (Hu et al., 2003). Therefore, research in the field of the office environment is critically important.

Several studies have analysed office chairs and different sitting positions by means of electromyography (Andersson and Ortengren, 1974; Gregory et al., 2006; Kingma and van Dieen, 2009; van Dieen et al., 2001), magnet resonance imaging (Baumgartner et al., 2012; Fryer et al., 2010; Zemp et al., 2013), leg/foot swelling evaluation (Bendix et al., 1985; Chester et al., 2002), motion tracking systems (Ellegast et al., 2012; Kingma and van Dieen, 2009; van Dieen et al., 2001), spinal shrinkage (Kingma and van Dieen, 2009; van Dieen et al., 2001), X-ray (Åkerblom, 1948; Schoberth and Hegemann, 1962), intra-disc pressure (Andersson and Ortengren, 1974; Rohlmann et al., 2001), subjective comfort/discomfort ratings (Carcone and Keir, 2007; Gregory et al., 2006; Groenesteijn et al., 2009), and pressure distribution measurements of the seat pan and the backrest (Carcone and Keir, 2007; Groenesteijn et al., 2009). However, there is still a lack of knowledge concerning the optimal office chair properties (surface, cushion properties, geometry, etc.) in order to prevent or reduce work-related musculoskeletal disorders caused by extended sitting periods. Pressure distribution measurements of the seat pan and the backrest are one of the most common objective methods to

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analyse or compare different chairs or sitting positions (Zemp et al., 2015) since pressure measuring systems are relatively cheap and easily applicable. For example Groenesteijn et al. (2009) evaluated a traditional office chair as well as a redesign of the same chair (more dense, 1 cm thicker and slightly more basin-shaped seat pan cushion) and analysed, among other things, the influence on the seat pan's peak pressure. They found no significant difference in the peak pressure between the two chairs. On the other hand, Carcone and Keir (2007) showed that the addition of a supplementary backrest to a standard chair is able to reduce peak and average pressure on the back in an upright posture by 35% and 20% respectively. However, according to the authors, further delineation between different objective seating parameters and comfort is still required in order to provide effective methods for understanding, reducing and preventing low back pain.

Stinson et al. (2003) analysed the pressure distribution of the seat pan of 63 subjects during sitting with different backrest reclination angles (0°, 10°, 20° and 30°). The study found that a 10° or 20° chair-recline did not significantly reduce the mean pressure, whereas a reclination of 30° had a significant effect. However, reclination angle had no significant impact on maximum pressure. Vos et al. (2006) assessed the pressure distribution of the seat pan of 24 subjects in order to investigate the influence of postural (trunk-thigh angle and use of armrest) and chair design differences (twelve office chairs). The authors concluded that chair design has the greatest effect on the pressure distribution on the seat pan, followed by participant effects and finally postural treatments (including different backrest angles and use of armrests). Pressure mats have also often been used to establish the interactions between seats and their users, since several studies indicate that pressure measures seem to be a highly associated and objective method for the quantification of subjective comfort/discomfort (De Looze et al., 2003; Mergl, 2006; Verver, 2004). However, the relationship between subjective comfort/discomfort and objective pressure measurements while sitting in office chairs remains to be demonstrated (Zemp et al., 2015).

There are currently several measurement systems for assessing the contact pressure between a chair and its user by means of resistive and capacitive sensor mats (Giacomozzi, 2010). Hochmann et al. (2002) evaluated four different modern seat pressure mapping systems (FSA, Xsensor, Tekscan ClinSeat, Novel Pliance) concerning, in particular, their accuracy, linearity and hysteresis. The relationship between the measured force (integration of all sensors) and the applied force (in this case an applicator with a diameter of 12 mm) did not increase strictly linearly for the Tekscan ClinSeat sensor mat, whereas the Novel Pliance mat showed almost a perfect linear characteristic curve. Furthermore, the study also revealed that the Novel Pliance sensor mat had the greatest accuracy out of the four analysed measuring systems. In terms of hysteresis, the systems of FSA and Novel showed almost no difference between the loading and unloading phase. The hysteresis errors were lower than 2% and 5% for the FSA and the Novel system, respectively. Further important source of errors – but not investigated in the described studies – are the bending effects of the pressure mat, which have also to be taken into account while interpreting the data of pressure mapping systems.

In order to be able to compare different office chairs or sitting positions by using pressure measurements, suitable outcome parameters have to be determined. Since many different parameters are used throughout the literature, there is little or no agreement in choosing the most appropriate ones. For example Carcone and Keir (2007) analysed the mean and peak pressure, the contact area as well as the centre of pressure for the office chair's seat pan and backrest. Others have examined parameters of pressure distribution such as the mass/force (Gutierrez et al., 2004), the

standard deviation of the pressure map (Gil-Agudo et al., 2009) and the peak, the circular as well as the transverse pressure gradients (Aissaoui et al., 2001; Hobson, 1992; Moes, 2007). Other studies looked at different regions of the human-chair contact area individually and analysed several pressure parameters for every segment (Zenk et al., 2012). In order to account for variability in this parameter, Vos et al. (2006) used principle component analysis to reduce the number of parameter domains and found that the mean and the peak seat pan pressure could be represented as a single parameter, which was able to explain 76.2% of the total variance. However, no studies exist that have examined the relationship between more than only the peak and mean pressures, as well as the different parameters' potential for evaluating and identifying different office chairs and sitting positions. Hochmann et al. (2002) emphasised that pressure mats are very sensitive to differences in the surface area properties of the product analysed. It would therefore appear to be important that sensor mats are calibrated for the corresponding surface. If the pressure values are not calibrated for a specific office chair, the calculated pressure parameters will be distorted. By calibrating the pressure distribution values, all pressure parameters, including parameters that describe the shape of the pressure distribution (e.g. pressure gradient), can be corrected.

The primary aim of this study was therefore to develop a chair-specific sensor mat calibration methodology in order to be able to compare the pressure distribution of different office chairs. Secondly, in order to allow effective and efficient analysis of sitting behaviour, we aimed to understand the inter-relationships and correlations between different pressure parameters during sitting. The final aim was to identify appropriate pressure parameters with the intention of being able to analyse and compare various office chairs appropriately.

2. Materials and methods

2.1. Chair-specific sensor mat calibration

As pressure mats are known to be influenced by the surface area, material properties as well as the geometry of the analysed product, the sensor mat of each seat pan was calibrated prior to subject measurements. Here, the pressure distribution of the buttocks of a sitting calibration dummy loaded at the centre of mass of the unloaded buttocks with different weights was analysed (5 s, 10 Hz) for each office chair three times (Fig. 1). The buttocks were modified with modelling clay and foam material as well as clothed with shorts in order to imitate the pressure mat response of a human buttock as closely as possible. The total weights of the calibration dummy were 4.40 kg, 6.41 kg, 9.30 kg, 14.20 kg, 24.30 kg and 44.20 kg in order to calibrate the pressure sensor mat for the full pressure range of our subject measurements. Attention was paid to ensure that the contact surface of the calibration dummy remained within the sensitive area of the pressure mat. The relationship between the mean pressure values was then assessed using the pressure on the mat measured using the calibration dummy with the different additional weights (p_{mat}) and the theoretical pressure value calculated using the known weights of the calibration dummy divided by the assessed contact area (p_{calib}). In order to determine each chair's pressure correction factor, f_{corr} , based on the assessed pressure (p_{mat}), f_{corr} was defined as p_{calib} divided by p_{mat} . Finally, x–y plots of the calibration measurements (p_{mat} against f_{corr}) clearly indicated an exponential function through the values of f_{corr} and p_{mat} with three parameters a_1 , a_2 and a_3 (1). This function was then used to correct the pressure values of the seat pan mat at each time point.



Fig. 1. Office chair with the pressure sensor mat on the backrest and the seat pan loaded with a calibration dummy's buttocks and plates (total weight: 24.30 kg) in order to calibrate the pressure mat for each office chair (top left), as well as a subject in the upright (top right), reclined (bottom left) and forward inclined sitting positions (bottom right).

$$f_{corr} = a_1 + a_2 e^{-\frac{p_{mat}}{a_3}} \quad (1)$$

2.2. Participants

Twenty healthy Swiss subjects (15 male, 5 female) with an average age of 26 years (range 20–37 years), an average height of 1.79 m (range 1.64–1.90 m; percentile: 33rd–90th of Swiss population (Bundesamt für Statistik, 2012)), an average weight of 73 kg (range 52–99 kg; percentile: 19th–89th) and an average BMI of

23 kg/m² (range 18–31 kg/m²; percentile: 4th–90th) took part in this study. Subjects with musculoskeletal disorders of the upper body as well as subjects who received medication for a prolonged period were excluded from the study. All subjects provided written informed consent prior to participation in this study, which was approved by the ethics commission of the ETH Zurich (no. EK 2011-N-40).

2.3. Experimental design

The Novel Pliance™ System (pliance-x 32 Expert Version

20.3.36, Novel GmbH, Munich, Germany) consisting of two sensor mats (16×16 sensors, $392 \times 392 \text{ mm}^2$) was used to analyse the pressure distribution of the seat pan and the backrest (Fig. 1) of nine different office chairs in a randomised order. The nine selected office chairs of six international manufacturers from various countries differed in movement mechanism, fabric and padding material, foam thickness and geometry as well as overall chair geometry in order to represent a broad spectrum of currently available products (Table 1). Prior to the measurements of each chair, subjects were instructed and assisted by the observer (based on the posture recommendations of the [Safety, Security, Health and Environment Staff Unit \(2010\)](#) of the ETH Zurich) to properly adjust the seat height, the seat depth, the backrest counter-pressure, the lumbar pad and the armrest, where applicable, to suitable and comfortable settings. The pressure distribution of the seat pan and the backrest was randomly assessed in an upright, reclined, and forward inclined sitting position (Fig. 1). In order to reproduce a real visual display unit (VDU) workplace situation the subjects were asked to place their fingers on the keyboard and to fix their eyes on the screen (Fig. 1). Further restrictions, regarding the sitting positions, were not made. After sitting for about one minute in the corresponding position, the subjects had to remain still during the measurement time of 5 s. The pressure distribution was assessed with a sampling frequency of 10 Hz, a resolution of 1 kPa and a measurement range between 0 kPa and 60 kPa. Eight parameters of pressure distribution were then calculated and compared in order to elucidate the inter-relationships and correlations between different pressure parameters during sitting.

2.4. Data analysis/pressure distribution parameters

Data analysis was performed using MATLAB (vR2010a, MathWorks Inc., Natick, USA). Firstly, each seat pan pressure parameter was corrected by multiplying by the corresponding chair pressure correction factor, f_{corr} . To analyse different pressure parameters, the mean pressure distribution over the whole measurement time of 5 s was calculated for the seat pan and the backrest. The following eight, most used static pressure distribution parameters in the literature (Zemp et al., 2015), were calculated for the seat pan (upright, reclined, forward inclined) and the backrest (reclined): Peak pressure (*PeakP*), mean pressure (*MeanP*), standard deviation of the pressure distribution (*STDP*), total contact area (*Area*), force (*Force*), maximum gradient (*MaxGrad*), mean gradient (*MeanGrad*) and the standard deviation of the gradient (*STDGrad*). The gradient (2) was defined by the geometrical addition of the pressure derivative of the two sensor mat directions (x, y) and resulted in a 15 by 15 matrix:

$$\text{Grad}(x, y) = \sqrt{\left(\frac{\partial P}{\partial x}\right)^2 + \left(\frac{\partial P}{\partial y}\right)^2} \quad (2)$$

Table 1
Office chair companies and the analysed office chair models in alphabetical order.

Company	Office chair model
HAG (Oslo, Norway)	Futu
Haworth (Holland, MI, USA)	Very task
Kinnarps (London, England)	9009
Knoll (New York, NY, USA)	Generation
Sato (Ebermannsdorf, Germany)	Galileo GlideTec
Vitra (Birsfelden, Switzerland)	ID Chair
Vitra (Birsfelden, Switzerland)	ID Chair Soft
Vitra (Birsfelden, Switzerland)	Prototype (confidential)
Wilkhahn (Bern, Switzerland)	ON Trimension

2.5. Statistical analysis

In order to quantify the interactions between the different pressure parameters, each of the eight analysed seat pan parameters were analysed by means of partial correlation analyses while controlling for the sitting position. Bivariate correlation analyses were made for the backrest parameters in the reclined sitting position. A two-tailed test of significance was carried out for both the seat pan and the backrest. In our context, an absolute correlation coefficient value $|R|$ of greater than 0.8 was considered as a high correlation between the two corresponding parameters. Scatterplot matrices containing the calculated Pearson's correlation coefficients were used to illustrate the correlation analyses of the seat pan and backrest pressure parameters. Box's M tests (Box, 1954) for the eight pressure parameters of the seat pan in the upright, reclined and forward inclined sitting positions as well as for the backrest in the reclined sitting position were carried out in order to examine the equality of covariance matrices for the different office chairs. Covariance matrices between groups were assumed to be equal for the purpose of a MANOVA, if the Box's M test p value was higher than 0.005 (Huberty and Petoskey, 2000). Furthermore, Kruskal–Wallis tests were carried out on the seat pan pressure parameters in the upright, reclined and forward inclined sitting positions, as well as on the backrest pressure parameters in the reclined position using the office chair as a grouping variable. If the Kruskal–Wallis test was significant, a Tamhane's T2 Post Hoc test was performed. Statistical significance was defined as $p < 0.05$ for all statistical analyses.

3. Results

3.1. Chair-specific sensor mat calibration

The parameters of the fitted exponential function for the nine office chairs can be found in Table 2. The calculated chair pressure correction factors, f_{corr} , varied between 1.88 (chair 3) and 5.67 (chair 7) for mean pressure values of 1 kPa. For values higher than 5 kPa, f_{corr} varied between 1.09 (chair 1) and 1.15 (chair 7; Fig. 2). As a result, our calibrated pressure values varied up to 300% for pressure values of 1 kPa and up to 5.5% for measured pressure values higher than 5 kPa between the different office chairs. Typically f_{corr} within each chair was highly dependent on the applied pressure and varied by a factor from 1.7 (chair 3) up to 4.9 (chair 7). The coefficients of determination R^2 were all higher than 0.912 with the exception of chair 7 ($R^2 = 0.820$; Table 2).

3.2. Connection between pressure parameters

Partial correlation analyses of the eight pressure parameters of the seat pan during subject sitting (Fig. 3) revealed absolute

Table 2
Calibration parameters a_1 , a_2 and a_3 of the fitted exponential function of the chair pressure correction factor, f_{corr} , for the nine office chairs, shown together with the coefficients of determination, R^2 , of the curve fit.

	a_1	a_2	a_3	R^2
Chair 1	1.087	6.977	0.508	0.951
Chair 2	1.109	48.263	0.287	0.957
Chair 3	1.100	4.509	0.571	0.976
Chair 4	1.125	50.589	0.287	0.982
Chair 5	1.131	90.858	0.249	0.969
Chair 6	1.092	263.400	0.210	0.917
Chair 7	1.151	11800.630	0.127	0.820
Chair 8	1.092	84.901	0.252	0.969
Chair 9	1.093	125.651	0.234	0.946

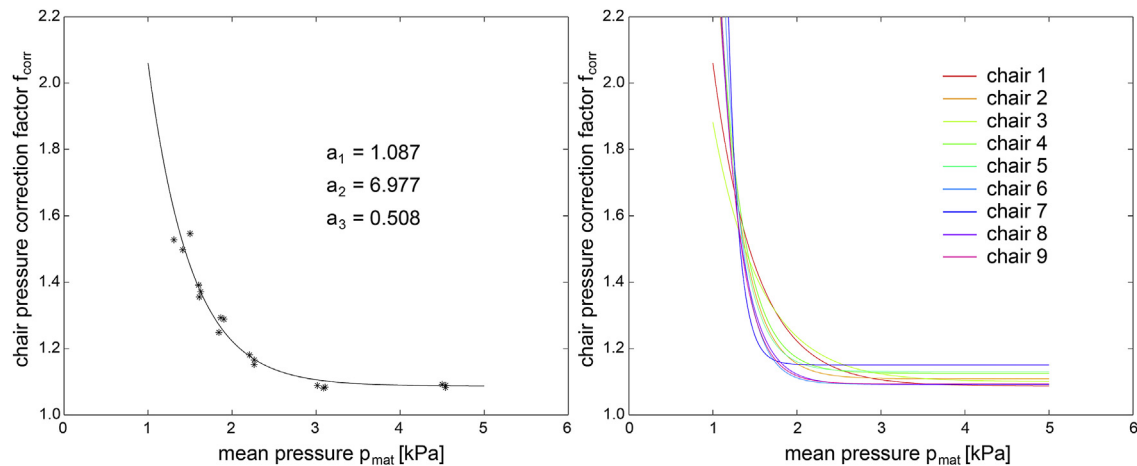


Fig. 2. Scatterplot and fitted exponential function of the assessed mean pressure, p_{mat} , and the chair pressure correction factor, f_{corr} , of the seat pan's pressure mat calibration exemplary for chair 1 (left) as well as the fitted curves of the nine different office chairs (right).

correlation coefficient values, $|R|$, of between 0.180 (MeanP and Area) and 0.932 (PeakP and STDP). Seven pairs of the seat pan pressure parameters showed a high correlation ($|R| > 0.8$). Correlation analyses of the eight pressure parameters of the backrest in the

reclined sitting position (Fig. 4) showed $|R|$ values of between 0.177 (Area and MaxGrad) and 0.921 (PeakP and MaxGrad). Again, seven pairs of the backrest pressure parameters showed a high correlation ($|R| > 0.8$). All the values of R of the seat pan and backrest

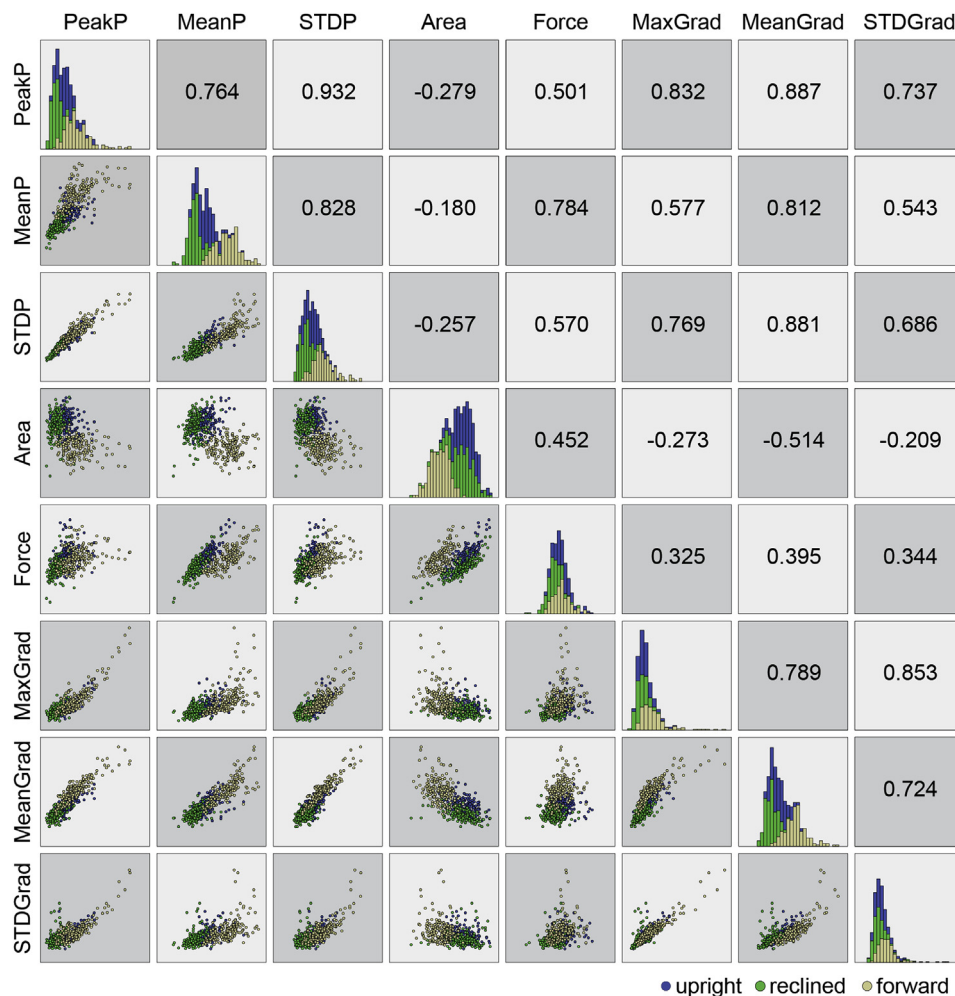


Fig. 3. Scatterplot matrix of the eight seat pan pressure parameters (bottom left) with the corresponding partial correlation coefficients, R , (top right) and the distribution of the eight parameters (diagonally downwards) in the upright (blue), reclined (green) and forward inclined sitting positions (yellow). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

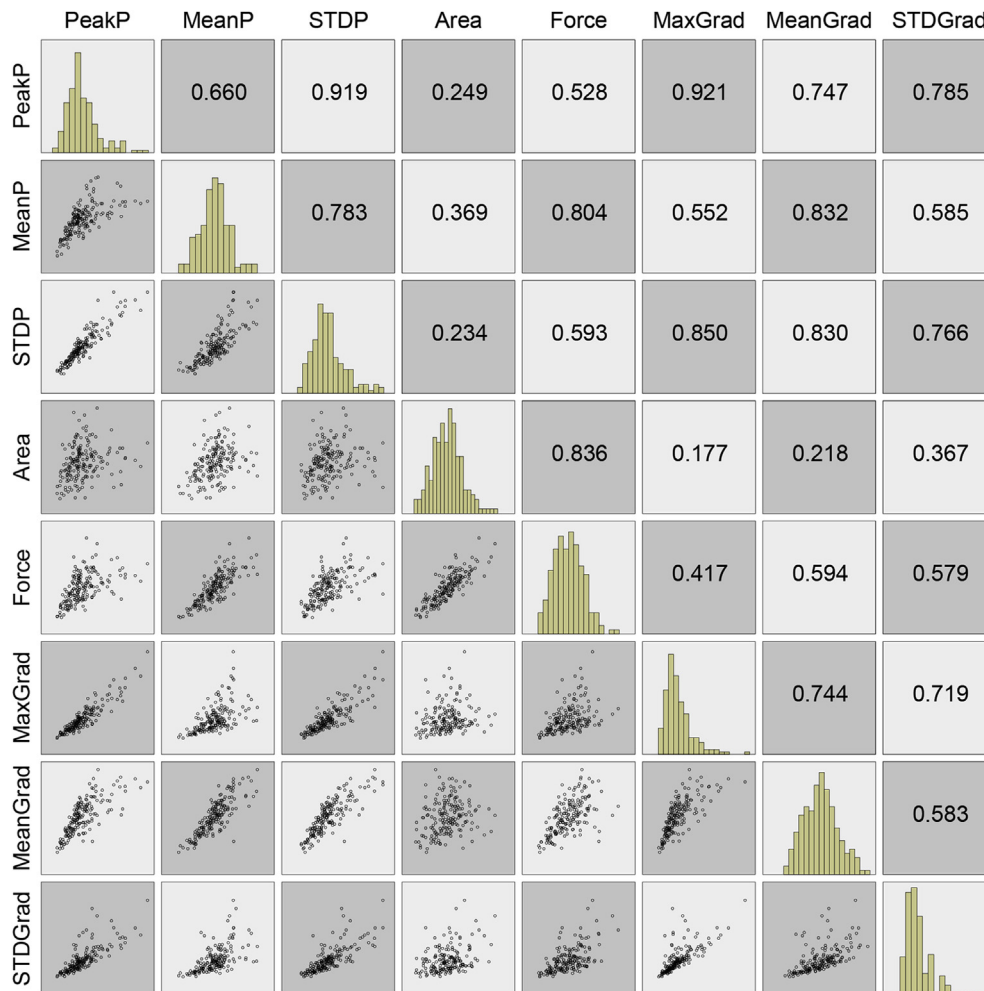


Fig. 4. Scatterplot matrix of the eight backrest pressure parameters in the reclined sitting position (bottom left) with the corresponding correlation coefficients, R , (top right) and the distribution of the eight parameters (diagonally downwards).

correlation analyses were significantly different from 0.

3.3. Pressure parameters of the different office chairs

Box's M test of equality of covariance matrices for the eight pressure parameters of the seat pan in three sitting positions as well as for the backrest in the reclined sitting position revealed that the covariance matrices for the different office chairs could not be assumed to be equal (p values < 0.001). Therefore, a MANOVA could not be performed on the data.

Kruskal–Wallis tests showed that there was at least one significant mean difference between the nine office chairs between the seat pan and backrest parameters as well as sitting positions, with the exception of the seat pan force in the upright ($p = 0.616$) and forward inclined ($p = 0.167$) sitting positions (Table 3).

The most significant differences in seat pressure parameters between pairs of office chairs could be found between the two parameters *MeanGrad* and *STDGrad* (Table 4), whereas the parameter *Force* revealed significant differences only between one pair of office chairs in the reclined sitting position. With 21 significant different differences between office chairs, the parameter *MaxGrad* showed the most significant differences for the backrest pressure distribution (Table 4).

4. Discussion

Working in an office chair is commonplace, and musculoskeletal disorders due to non-ergonomic sitting positions or prolonged sitting postures are therefore ubiquitous (Naqvi, 1994; Winkel and Jorgensen, 1986). Measurement systems for assessing the pressure distribution of the seat pan and the backrest are often used to quantify sitting comfort or discomfort in an objective manner in order to reduce or prevent low back pain (Zemp et al., 2015). However, there is little agreement in the literature for selecting the most appropriate parameters and especially for a reliable comparison of different office chairs (Aissaoui et al., 2001; Carcone and Keir, 2007; Gil-Agudo et al., 2009; Gutierrez et al., 2004).

4.1. Chair-specific sensor mat calibration

Due to the fact that pressure sensor mats are highly sensitive to the different surface, as well as properties of the seat pan and the backrest of the office chair (Hochmann et al., 2002), a chair-specific sensor mat calibration is critically important. Our measurements using the sitting calibration dummy support the findings of Hochmann et al. (2002), who demonstrated that pressure mats are highly influenced by the seat's properties. With R^2 values between 0.820 and 0.982, the fitted exponential function was clearly suitable for describing the relationship between the measured mean

Table 3

P values of the Kruskal–Wallis tests with the nine office chairs as the grouping variable for the eight pressure parameters of the seat pan (upright, reclined and forward inclined sitting positions) as well as of the backrest (reclined sitting position).

		PeakP	MeanP	STDP	Area	Force	MaxGrad	MeanGrad	STDGrad
Seat pan	Upright	<0.001	<0.001	<0.001	0.019	0.616	<0.001	<0.001	<0.001
	Reclined	<0.001	<0.001	<0.001	<0.001	0.015	<0.001	<0.001	<0.001
	Forward	0.043	0.003	0.014	<0.001	0.167	<0.001	<0.001	<0.001
Backrest	Reclined	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001

pressure value p_{mat} and the chair pressure correction factor f_{corr} . Due to the fact that f_{corr} was not constant for every measured pressure value, the chair-specific sensor mat calibration seems to play an important role for the calculation of the *PeakP*, *MeanP*, *STDP* and *Force* but also for the shape of the pressure distribution explained by the gradient parameters *MaxGrad*, *MeanGrad* and *STDGrad*. These results demonstrate that a chair-specific calibration is required before the pressure distribution of different office chairs can be compared. To our knowledge, there is no current study that has calibrated pressure sensor mats on a chair-specific basis. It is quite conceivable that reported differences in the pressure distribution between different office chairs are due to the influence of the surface area properties on the pressure measurements. The opposite argument could also apply, that significant differences in pressure parameters might have been found if the pressure mats had been calibrated in the previous studies.

4.2. Connection between pressure parameters

When quantifying the pressure distribution with various pressure parameters, it is an inevitable consequence that some parameters are interconnected with one another. It is conceivable, for example, that the parameters *PeakP* and *STDP* are correlated with each other, since high *PeakP* values involve a greater range of pressure and therefore result in a higher *STDP*. We have performed partial correlation analyses in order to find pressure parameter pairs with high correlations. Such parameters pairs contain similar information and can therefore be reduced to only one without losing important information regarding the pressure distribution.

The results showed that pressure distributions on the seat pan can be described by using a set of only four parameters *Area*, *Force*, *MaxGrad* and *MeanGrad*. The *Force* and the *Area* of the seat pan pressure distribution showed no high correlations ($R < 0.8$) to other pressure parameters and could therefore not be explained using other parameters, whereas *MaxGrad* is closely correlated to (and therefore describes) *PeakP* and *STDGrad*, while *MeanGrad* characterises the parameters *PeakP*, *MeanP* and *STDP*. In other words, the characteristics of the pressure distribution of the seat pan could be described using only four parameters (*Area*, *Force*, *MaxGrad*, *MeanGrad*), while the other parameters (*PeakP*, *MeanP*, *STDP*, *STDGrad*) could not provide added value, since they are highly connected to the parameters *Area*, *Force*, *MaxGrad* and/or *MeanGrad* of the seat pan pressure distribution. Based on the correlation analyses of the backrest pressure distribution, all backrest pressure

parameters could be characterised by a set of three parameters *STDP*, *Force* and *STDGrad*. *PeakP*, *MaxGrad* and *MeanGrad* could be explained by the parameters *STDP* and *MeanP*, and *Area* by the parameter *Force*, while *STDGrad* showed no high correlations to other parameters. As a result, the parameters of the pressure distribution of the seat pan as well as of the backrest in this study could be reduced from eight to four and to three, respectively, without a significant loss of information.

4.3. Pressure parameters of the different office chairs

In order to find the most meaningful and appropriate pressure parameters, the pressure distributions of the different office chairs have been statistically compared. Studies analysing the pressure distribution of the seat pan in subjects with the intention to find significant parameter differences between different office chairs should use the three gradient parameters (*MaxGrad*, *MeanGrad*, *STDGrad*), since the most significant different chair pairs could be found with these parameters. The most unsuitable parameter for the seat pan in order to differentiate pressure distribution measures between office chairs is *Force*, which was only significantly different between one chair pair.

All backrest parameters with the exception of the parameter *Area* could be used to differentiate between the pressure distributions of the different office chairs. However, it has to be taken into account that backrest parameters are only suitable for reclined sitting positions, since in the forward inclined as well as the upright sitting positions the subjects did not necessarily have full contact with the backrest.

It remains unclear whether the differences in the seat pan and backrest pressure parameters are between seats, between the adjustments of the chair, or between the specific posture each person took in the corresponding position (upright, reclined, forward inclined). However, due to the fact that the subjects adopted a conventional VDU working position (fingers on keyboard, eyes fixed on screen) with the most comfortable chair adjustments, the three gradient parameters for the seat pan and all eight backrest parameters (except *Area*) are suitable for describing and quantifying the pressure distribution of a specific chair in a VDU working position.

4.4. Limitations

One of the main limitations of the current study is that only static pressure parameters were evaluated. Thus, the current

Table 4

Number of significant different office chair pairs (Tamhane's T2 Post Hoc test) of the different pressure parameters of the seat pan (upright, reclined and forward inclined sitting positions) as well as of the backrest (reclined sitting position).

		PeakP	MeanP	STDP	Area	Force	MaxGrad	MeanGrad	STDGrad
Seat pan	Upright	6	5	3	0	x	9	19	22
	Reclined	7	10	8	5	1	8	20	11
	Forward	0	2	1	5	x	4	6	10
	All	13	17	12	10	1	21	45	43
Backrest	Reclined	14	16	17	4	13	12	21	10

x: Tamhane's T2 Post Hoc test was not carried out due to a non-significant Kruskal–Wallis test for the corresponding pressure parameter and sitting position.

results cannot be transferred into or compared against dynamic measures, where the pressure distribution is analysed over time. Crawford et al. (2005) recommend recording the pressure distribution after four minutes of sitting, since the pressure parameters increase considerably in the first four minutes. In our study, recording was started after only one minute of sitting in the corresponding position. However, as we paid attention that the measurements were all taken at the same time point after sitting down for every subject, sitting position as well as office chair, the influence of the increase in the pressure in the first four minutes for our analysed parameters could, at least to a certain extent, be controlled.

Since Kolich and Taboun (2004) as well as Hostens et al. (2001) showed that static pressure distribution measures in automobile seats and in mobile agricultural machinery seats are repeatable, no additional repeated measurements were performed in the subjects in our study – leaving the repeatability of the analysed parameters including the adopted sitting positions somewhat unknown. Here, it should be noted that the requirements and the environment in vehicle seats are certainly different than those in office chairs, since special safety standards and boundary conditions (e.g. seat belts, hand placement on steering wheel, feet on pedals etc.) are required to be fulfilled (Lueder, 1983; Zemp et al., 2015). As a result, the analysis of sitting repetitions in office chairs remains a clear requirement for further understanding of seating posture and sitting behaviour and should be undertaken as a priority in future studies.

5. Conclusions

Pressure distribution measurements with different office chairs are strongly influenced by different material properties and geometry of the padding material and the pressure mat should therefore be calibrated for every chair. Partial correlation analyses suggested that of the eight analysed pressure parameters of the seat pan and the backrest, a set of only four (seat pan) or three (backrest) parameters would be sufficient to analyse subjects' sitting and postural characteristics. In order to find significant differences between different office chairs, gradient parameters should be analysed for the seat pan, whereas for the backrest all parameters are suitable with the exception of the parameter *Area*. However, it should be noted that for specific research questions, meaningful and relevant pressure parameters have to be selected individually. For instance, future studies are needed to analyse and compare pressure distribution parameters in relation to musculoskeletal complaints in the office environment.

The current study has introduced a novel chair-specific sensor mat calibration in order to make pressure distribution measures of different office chairs comparable. Furthermore, the evaluation of the different seat pan and backrest pressure parameters provide assistance for the selection of meaningful and appropriate pressure distribution parameters for future studies.

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